Example Exam Questions from Fall 2020

True or False

1. T ____ Sequential search is an example of an algorithm that executes in O(n) time (worst case time).
2. F ____ "If the dough is too sticky, add a quarter cup of flour" is an example of an iterative statement.
3. T ____ Big O notation is used to describe an algorithm's performance/execution time.
4. T ____ One benefit of two's complement over sign magnitude notation is that addition works the same as with unsigned numbers.
5. T ____ The number 238 is a valid number in octal (base 8).
6. F ____ A 2-input OR gate is only true when both inputs are true.
7. T ____ A 4-input AND gate is false when only one input is true.
8. F ____ A USB flash drive is an example of volatile storage.
9. F ____ The ALU is responsible for controlling input and output devices, such as the keyboard.
10. F ____ The assembler converts an instruction from machine language to assembly language.

Fill in the Blank

A-10
B-11
C-12

1. A byte is a collection of eight bits.
2. To represent the number 13 in hexadecimal notation, we use the letter D.
3. The minimum value in 8 bit two's complement is 127 and the maximum value is -128.
4. Converting an analog signal (e.g., sound) to digital data is called digitizing.
5. The Cache is small and fast general-purpose memory on the CPU.
6. RAM stands for random access memory
7. high level are popular to program in because they are easier to read and write than assembly.
8. When using scientific notation to represent a fractional number, in the equation: $\pm M \times B^{\pm E}$, "M" stands for mantisa, "B" stands for base, and "E" stands for exponent.

Multiple Choice

1. $O(n^3 * n^2 + n^4)$ simplifies to:
    a. $O(n^3)$
    b. $O(n^4)$
    c. $O(n^5)$
    d. $O(n^5 + n^4)$
    e. None of the above

2. Which of the following is/are types of operations used to construct an algorithm?
    a. Textual
    b. Digital
    c. Sequential
    d. All of the above
    e. None of the above

3. When we are looking at an algorithm's efficiency to be run on an embedded device, we care about:
    a. Time efficiency
    b. Space efficiency
    c. Power efficiency
    d. Both a and b
    e. None of the above

4. Imagine a number given in base 12. Which digits or letters would be invalid in such a system (select all that apply)?
    a. 0
    b. 2
    c. 9
    d. A
    e. D

5. How would you represent -1 in 4 bits of binary?
    a. 1111
    b. 1001
    c. 0001
    d. 0111
    e. Not enough information provided to know

6. Which equation gives the maximum unsigned integer representable by n bits?
    a. 2n+1
    b. 2n−1
    c. 2^n+1
    d. 2^n-1
    e. None of the above

7. What is a tautology in computer science?
    a. A way to describe gate connections
    b. A Boolean expression evaluated to a known value
    c. A Boolean expression which is always true
    d. A computer science discipline about computer logic
    e. None of the above

8. Which of the following is not a basic Boolean operation?
    a. OR
    b. NOT
    c. AND
    d. FOR
    e. None of the above

9. A 3 input AND gate has eight input combinations. How many of these will result in a true output?
    a. 1
    b. 2
    c. 5
    d. 8
    e. None of the above

10. Choose the correct order of access speeds, from fastest to slowest.
    a. Cache, Drive, Memory
    b. Memory, Drive, Cache
    c. Cache, Memory, Drive
    d. Drive, Memory, Cache
    e. Memory, Cache, Drive

11. Suppose memory is addressable with 4 bits. How many addresses exist?
    a. 4
    b. 8
    c. 16
    d. 32
    e. None of the above

2^4

12. In this assembly language instruction, which part is the opcode?
    Decrease: Input M[40] -- Put value into memory address 40
    a. Decrease:
    b. Input
    c. M[40]
    d. -- Put value into memory address 40
    e. None of the above

13. Which of these represents code using binary?
    a. Two's complement
    b. Machine language
    c. Assembly language
    d. Natural language
    e. None of the above

Short Answer Questions

1. According to the institution's website, Mines has 6,268 students. Your resourceful 101 instructors managed to put together a manual of all Mines students, containing their CWID and full name. The instructors sorted the manual numerically based on CWIDs. You can leave your answer in equation form or simplify - your choice.
   a. In the worst case, how many comparisons would you have to make in order to find "Rena Loving" in the manual?

   $O(n) = 6,268$

   b. In the worst case, how many comparisons would you have to make in order to determine if the CWID 10804825 belongs to a student on campus?

   $\log_2(6268) = 13$

2. Given the list of numbers:
   [1, 3, 5, 7, 9, 10, 12, 15, 18, 21, 25, 28, 30, 33, 35, 40]
   a. When using a sequential search, how many comparisons are needed to find 21?

   10

   b. When using a binary search, what comparisons are needed to find 21?
   (write out the numbers that you would compare with 21)

   15, 28, 21 → 3

3. Imagine using a computer with only 4 bits available. What happens when you add 1100 and 1000 (in unsigned binary)?

   1100
   1000

   10100 → 0100

   overflow

4. The following 8 bit <u>two's complement</u> numbers are given in hexadecimal:
   72 + 4A
   Add them together and report the result in binary and decimal form.

   7 : 0111 → 01110010
   2 : 0010 +
   4 : 0100 → 01000010
   A : 1010 10111100

   A-11

   = -68

5. A drive has 100 GiB storage. How many 750 MiB videos can be stored on that drive, assuming it's empty to begin with? Assuming base 2 definitions, GiB = 2^{30} B and MiB = 2^{20} B

$100\,GiB \cdot \frac{2^{10}\,MiB}{1\,GiB} = \frac{MiB}{750\,MiB} = 136$ photos

6. Represent the decimal number 56.625 in binary and normalize the result.

$32+16+8+\frac{1}{2}+\frac{1}{8}$ $\quad 111000.101 \Rightarrow 1.11000101 \times 2^5$

7. Provide the truth table for the NAND gate (pictured right).

a	b	out
0	0	1
0	1	1
1	0	1
1	1	0



8. Convert the decimal value 303_{10} to:
   a. octal (base 8)

   b. hexadecimal

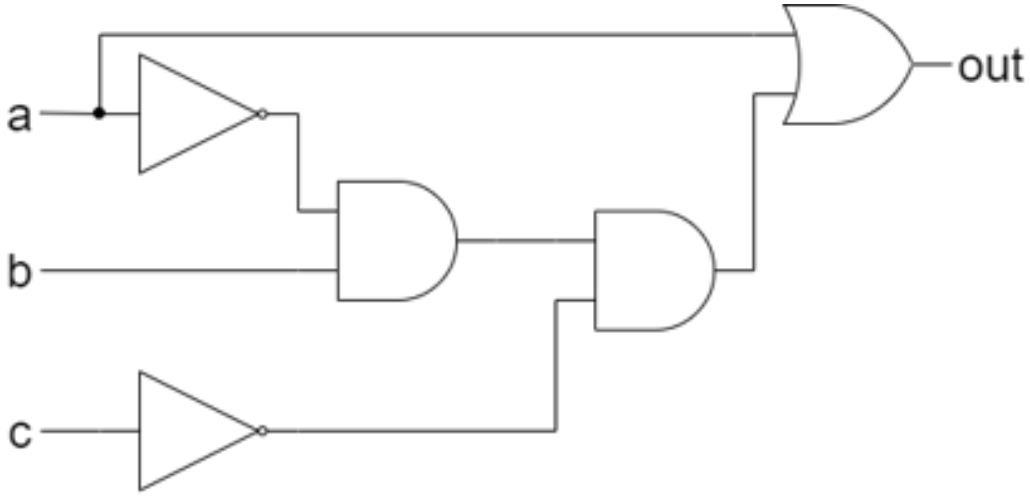
9. Suppose your instructor wrote the following pseudocode for an algorithm, which converts an unsigned binary number into decimal. What is wrong with the pseudocode and what would you do to fix it?

```
Initialize sum to 0
Initialize index to 0
Initialize bit_number to 0
While index is less than the total number of bits:
        Set bit_number equal to the bit at index
        If bit_number is 1:
                Add 2^bit_number to sum
        Increment index
Print the sum
```

Long answer.

Work through each problem carefully.

1. Write the Boolean expression and create a truth table for the following circuit:



2. Write an algorithm in pseudocode that takes an unsorted list called my_list and outputs the smallest number in that list without the use of built-in functions.

3. For the following question, use the following assembly instructions:
   ADD, SUBTRACT, INPUT, OUTPUT, STOP.

   Convert this short pseudocode program to assembly. Assume variable *a* is assigned to M[12] and variable *b* is assigned to M[14].

   1. Get *a* from user
   2. Get *b* from user
   3. Set *a* = *a* + *a* + *b*
   4. Set *b* = *b* + *b*
   5. Print *a*
   6. Print *b*
   7. End